

Abdullah Khaled

📍 Frisco, TX 🌐 akhaled247.github.io [in](#) [akhaled247](#) [🔊](#) [akhaled247](#)

Education

- Wakeland High School**, High School Diploma – Frisco, TX Aug 2023 – May 2027
- Unweighted GPA: 4.0
 - 18 AP courses by graduation (completed and planned)
- Frisco ISD Career and Technical Education Center**, Project Lead The Way – Frisco, TX Aug 2023 – May 2027
- Frisco ISD Career and Technical Education Center Project Lead the Way (PLTW) Nationally Distinguished Engineering Program
 - Courses: PLTW Introduction to Engineering Design (9th grade), PLTW Principles of Engineering (10th grade), PLTW Digital Electronics (11th grade), PLTW Engineering Design and Development (12th grade)
 - Engineering Portfolio: akhaled247.github.io/pltw-engineering

Technical Experience

- Chief Technical Advisor**, ITKAN Robotics (FRC 9128) – Frisco, TX Aug 2025 – present
- Developed components of competition robot software stack integrating localization, path planning, telemetry, and mechanism control in Java
 - Built publish-subscribe telemetry/logging framework and finite-state-machine path planning (Dijkstra on time-weighted graphs) for real-time robot control and diagnostics
 - Led development of robot state estimation and localization using sensor fusion of AprilTag vision and swerve-drive odometry, achieving <2 in positional error
- Software Lead**, ITKAN Robotics (FRC 9128) – Frisco, TX Dec 2024 – July 2025
- Provided technical leadership on organization of software projects and mentored new programming team members
 - Developed self-paced robotics programming curriculum and training website used by 20+ FRC and FTC students and reaching 1000+ impressions worldwide (mantik.netlify.app)
 - Robot codebase and documentation: akhaled247.github.io/first/reefscape
- Programming Mentor**, ITKAN Robotics of ICF (FTC 26357, 28931) – Frisco, TX Sept 2024 – Feb 2026
- Implemented Kalman-filtered state estimation combining AprilTag vision and wheel odometry for real-time robot localization and automated turret targeting
 - Mentor refugee students in FLL workshops and lead weekly summer training for FTC and FRC students using self-developed OARobotics curriculum (akhaled247.github.io/personal/oarobotics)
- Co-Founder, President**, Wakeland High School Wolverine Robotics (FTC 33791) – Frisco, TX Aug 2024 – present
- Co-founded Wakeland robotics club, growing membership from 0 to 50+ students in the first year
 - Secured funding and constructed competition-ready robot in under 2 weeks
 - Teach CAD software (Onshape) and Java programming through monthly technical workshops
 - Manage club operations including mission development, curriculum design, and budget management
- Math and Reading Instructor**, Gideon Math and Reading Center – Frisco, TX June 2024 – present
- Instruct K-11th grade students in mathematics (Algebra, Geometry), reading, and grammar
 - Analyze student performance data to identify learning gaps and develop personalized plans
- Math and Reading Teaching Assistant**, Gideon Math and Reading Center – Frisco, TX Jan 2022 – May 2024
- Graded K-11th work (including Algebra and Geometry) and supported students in completing assignments

Projects

OralVision – The Conrad Challenge

Oct 2025 – present

- Designed oral disease detection system using HIPAA-compliant federated learning with TensorFlow Federated, enabling distributed model training without sharing patient data
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs
- Conrad Challenge Innovator Award and Alternate Finalist; Diamond Challenge Semi-Finalist
- Patent application in progress with the United States Patent and Trademark Office (USPTO)

WatchFall – Samsung Solve for Tomorrow

Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Engineered hybrid dataset combining public and custom annotated data, reducing false positive rate

Skills

Software & Data: Python, Java, JavaScript, R, Google Sheets

Robotics, ML & Vision: TensorFlow, TensorFlow Federated, OpenCV, SolvePnP, Kalman Filtering, Finite State Machines, Mechanism Simulation, Control Theory, Federated Learning

CAD, Electronics & Prototyping: Fusion 360, Onshape, NI Multisim, KiCAD, EasyEDA, Soldering

Web & Productivity: HTML, CSS, Jekyll, Google Workspace, Microsoft Office

Leadership & Activities

Vice President, Mu Alpha Theta/Math National Honor Society

Aug 2024 – present

- Lead organization as Vice President, managing budget for 100+ members
- Expanded Math Lab Program and coordinated peer tutoring, attaining over 20% member participation
- Organized annual middle school math competition with 100+ participants, 3 participating schools, and 35+ volunteers to promote STEM

Treasurer, Mu Alpha Theta/Math National Honor Society

Aug 2023 – July 2024

- Managed organization budget for 100+ members
- Designed and implemented Math Lab Program coordinating student-led tutoring in Algebra I courses with teachers

Co-Founder and President, Muslim Student Association (MSA) – Wakeland High School, Frisco, TX

Sept 2024 – present

- Developed mission statement, recruited teacher sponsor and members, and presented activities to school administration
- Negotiated with school administration to establish a dedicated prayer room, coordinating facility setup and policies

Lead Math and English Instructor, Ma'ruf Project Taleem – Dallas, TX

Sept 2023 – present

- Teach Math and English to refugee students and mentor them in academics, time management, and social integration
- Coordinate small-group instruction and parent communication to support students' long-term academic progress

Honors & Awards

- (2026) OralVision: Conrad Challenge Innovator Award & Alternate Finalist; Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner; Control Award Winner (FiT-North Dallas Semi-Regional, 25 teams; FiT-North U-League)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist; Event Winner (FiT Victoria, 25); Autonomous Award (FiT Plano, 31); Creativity Award (FiT Amarillo, 36)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) Collegeboard: AP Scholar with Distinction; National Recognition Program School Recognition Award